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Coordenadas y cambio de variables

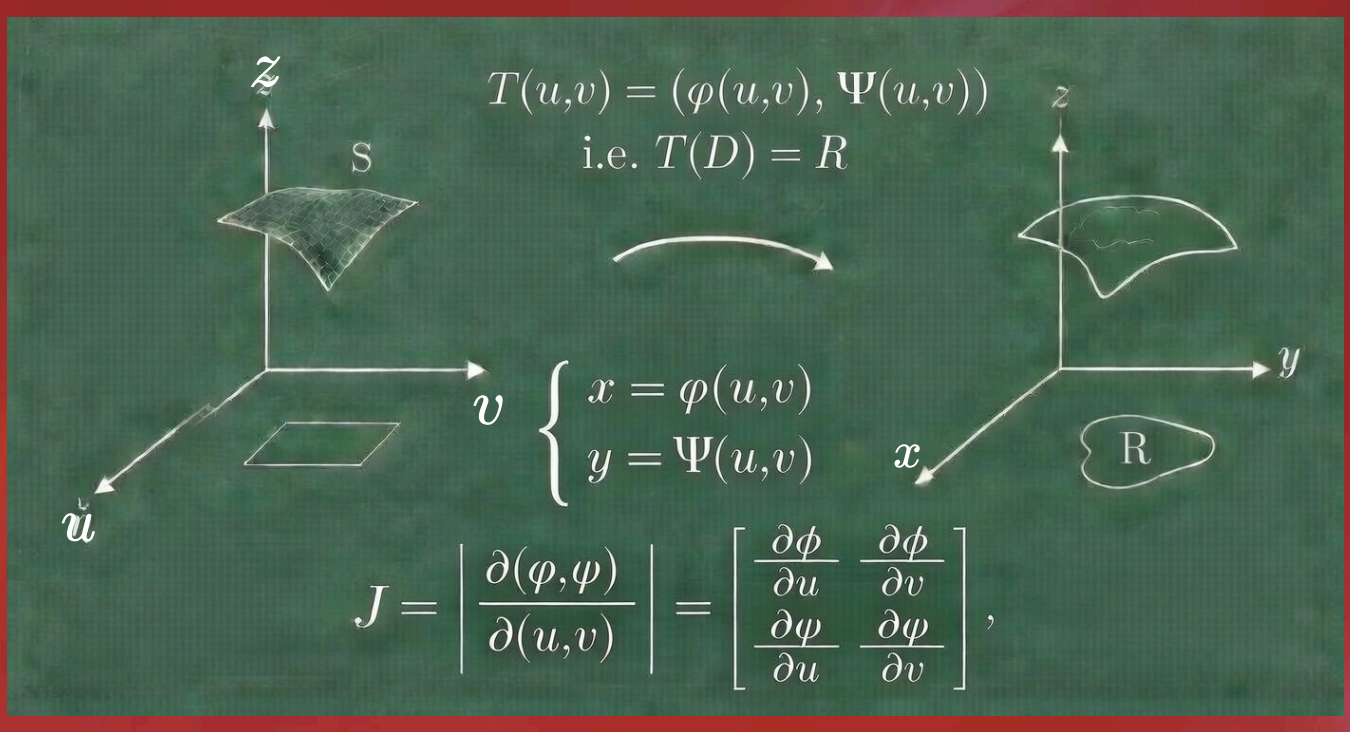
1. Teorema de cambio de variable

Sean $x = \phi(u,v), y = \psi(u,v)$ funciones diferenciables y con derivadas continuas. Entonces

$$\iint_D f(x,y)dydx = \iint_R f(\phi(u,v),\psi(u,v))|J|dvdu$$

donde

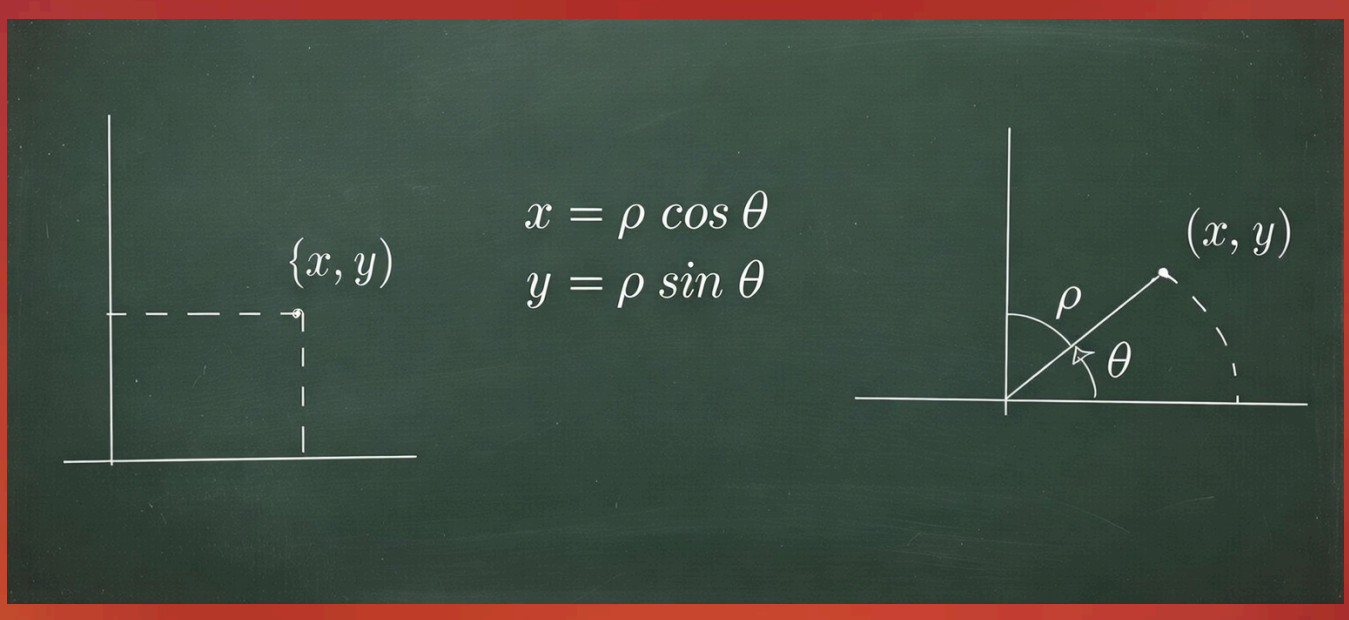
$$J = \left|\frac{\partial(\phi,\psi)}{\partial(u,v)}\right| = \begin{vmatrix} \frac{\partial\phi}{\partial u} & \frac{\partial\phi}{\partial v} \\ \frac{\partial\psi}{\partial u} & \frac{\partial\psi}{\partial v} \end{vmatrix}, f(D) = R.$$



2. Coordenadas polares

$x = \rho\cos\theta$ entonces $J = \rho, \rho \neq 0,$

$y = \rho\sin\theta$

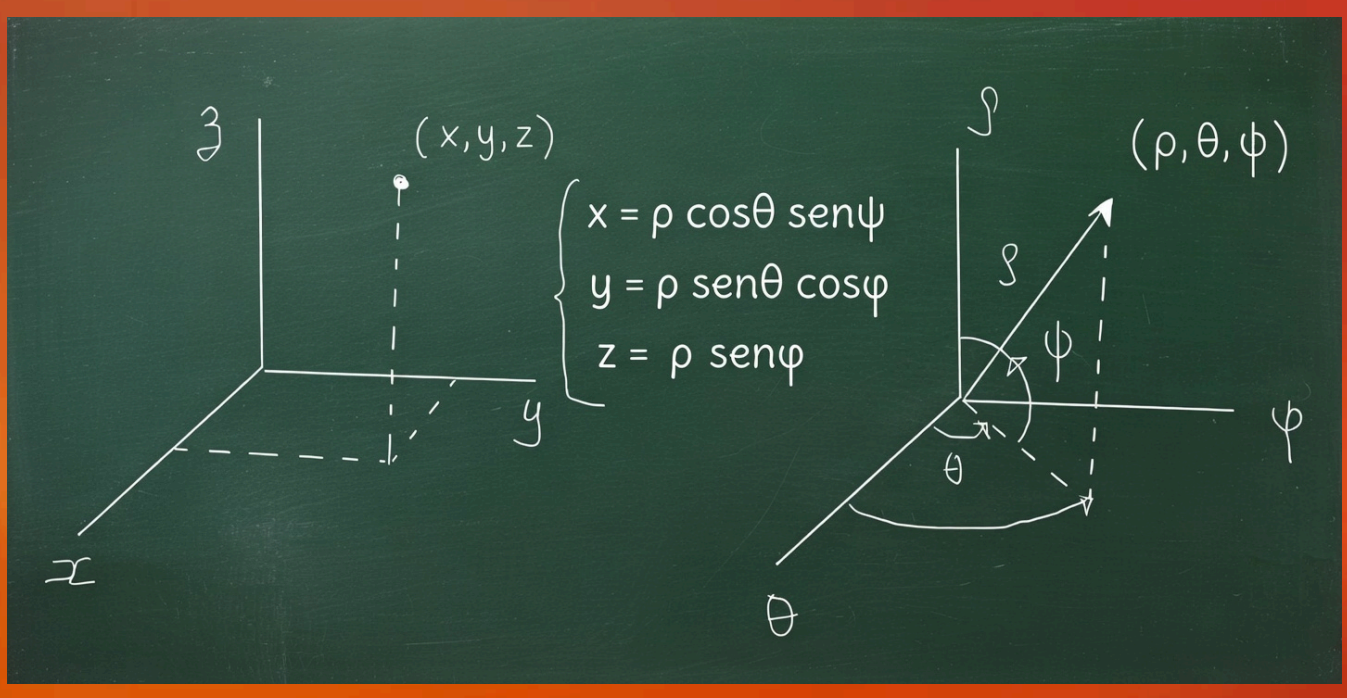


3. Coordenadas esféricas

$x = \rho\cos\theta\cos\varphi$

$y = \rho\sin\theta\cos\varphi$ entonces $J = \rho^2\cos\varphi, \rho \neq 0$

$z = \rho\sin\varphi$ $\theta \in [o, 2\pi), \varphi \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$



INFOGRAFÍA DEL TEOREMA DE CAMBIO DE VARIABLES EN INTEGRACIÓN DE VARIAS VARIABLES